

PROJECT REPORT

Census Income Analysis and Prediction

Exploratory Data Analysis and Machine Learning Classification of the UCI Adult / Census Income Dataset

Prepared using Python, Pandas, Scikit-learn, and Google Colab

Dataset: census-income.csv (32,561 records, 15 attributes)

July 2026

Table of Contents

1. Executive Summary
2. Introduction and Objectives
3. Dataset Description
4. Data Preprocessing
5. Data Manipulation and Descriptive Analysis
6. Exploratory Data Analysis (EDA)
7. Machine Learning Methodology
8. Linear Regression: Education vs. Hours Worked
9. Logistic Regression: Simple Model (Occupation Only)
10. Logistic Regression: Multiple Model (Age, Workclass, Education)
11. Decision Tree Model
12. Random Forest Model (300 Trees)
13. Model Comparison and Discussion
14. Conclusion
15. Limitations and Future Scope
16. References and Tools

1. Executive Summary

This project performs an end-to-end data science workflow on the Census Income (“Adult”) dataset, a well-known benchmark dataset containing demographic and employment information for 32,561 individuals from the 1994 US Census database. The objective is twofold: first, to clean and explore the dataset to understand the relationships between demographic attributes and income level; and second, to build and compare multiple machine learning models that predict whether an individual earns more than \$50,000 per year.

After removing incomplete records (rows containing missing values encoded as “?”), the dataset was reduced from 32,561 to 30,162 clean records. Exploratory analysis revealed clear relationships between income and factors such as education level, hours worked per week, marital status, and occupation. Four predictive models were built: a simple Logistic Regression using occupation alone, a multiple Logistic Regression using age, workclass, and education, a Decision Tree using all available features, and a Random Forest (300 trees) using all available features. A Linear Regression model was also built to predict weekly working hours from education level.

The Random Forest model achieved the highest classification accuracy at 85.41%, followed by the Decision Tree at 81.09%, the multiple Logistic Regression at 78.52%, and the simple Logistic Regression at 75.11%. These results confirm the well-established finding that ensemble tree-based methods generally outperform simpler linear models on this type of structured, mixed-type tabular data.

2. Introduction and Objectives

2.1 Background

Predicting income level from demographic and employment attributes is a classic problem in applied machine learning, frequently used to benchmark classification algorithms and to illustrate real-world issues such as missing data, categorical encoding, and class imbalance. The Census Income dataset used in this project is derived from the 1994 US Census Bureau database and has become a standard reference dataset in the machine learning community.

2.2 Objectives

- Clean and preprocess the raw census data by identifying and removing missing values.
- Perform structured data manipulation — slicing, filtering, grouping, and aggregating — to answer specific business-style questions about the population.
- Conduct exploratory data analysis (EDA) with visualizations to uncover patterns and relationships among features.
- Build a Linear Regression model to study the relationship between education and weekly working hours.

- Build and evaluate Logistic Regression, Decision Tree, and Random Forest classifiers to predict whether an individual's annual income exceeds \$50,000.
- Compare model performance and draw conclusions about which approach is best suited to this dataset.

2.3 Tools and Technologies

- Python 3 (Google Colab / Jupyter Notebook environment)
- Pandas and NumPy — data manipulation
- Matplotlib and Seaborn — data visualization
- Scikit-learn — machine learning model building and evaluation

3. Dataset Description

The dataset consists of 32,561 rows and 15 columns. Each row represents one individual, and the target column 'annual_income' indicates whether that person earns ' $\leq 50K$ ' or ' $> 50K$ ' per year. The table below describes each attribute.

#	Column	Type	Description
1	age	Numeric	Age of the individual in years
2	workclass	Categorical	Type of employer (Private, Self-emp, Government, etc.)
3	fnlwgt	Numeric	Census sampling weight
4	education	Categorical	Highest level of education attained
5	education-num	Numeric	Education level encoded as a number
6	marital-status	Categorical	Marital status of the individual
7	occupation	Categorical	Type of occupation
8	relationship	Categorical	Relationship role within the household
9	race	Categorical	Race of the individual
10	sex	Categorical	Gender (Male / Female)
11	capital-gain	Numeric	Income from investment sources
12	capital-loss	Numeric	Losses from investment sources
13	hours-per-week	Numeric	Number of hours worked per week
14	native-country	Categorical	Country of origin
15	annual_income	Categorical (Target)	Income class: $\leq 50K$ or $> 50K$

Key summary statistics for the numeric columns (before cleaning):

Statistic	age	education-num	capital-gain	hours-per-week
Mean	38.58	10.08	1077.65	40.44
Std. Dev.	13.64	2.57	7385.29	12.35
Min	17	1	0	1
Median	37	10	0	40
Max	90	16	99999	99

4. Data Preprocessing

Real-world datasets rarely arrive in a perfectly clean state. In this dataset, missing values were not left blank; instead, they were encoded using the '?' character, primarily in the workclass, occupation, and native-country columns.

4.1 Identifying Missing Values

All occurrences of '?' were replaced with a proper NA / NaN marker so that pandas could correctly recognize them as missing. The count of missing values by column was:

Column	Missing Count
workclass	1,836
occupation	1,843
native-country	583
All other columns	0

4.2 Removing Missing Values

All rows containing at least one missing value were removed from the dataset using a listwise deletion approach. This reduced the dataset from 32,561 rows to 30,162 rows — a loss of about 7.4% of the records, which is an acceptable trade-off for ensuring data quality in downstream analysis and modeling.

5. Data Manipulation and Descriptive Analysis

This section answers a series of structured queries on the cleaned dataset, demonstrating common data manipulation operations — column selection, filtering, sampling, grouping, and aggregation.

5.1 Extracting the 'education' Column

The 'education' column was extracted independently into a variable named census_ed for isolated analysis of educational attainment.

5.2 Extracting Columns from 'age' to 'relationship'

The first eight columns of the dataset (age through relationship) were extracted together into census_seq, representing the demographic block of attributes.

5.3 Extracting Columns 5, 8, and 11

Columns number 5, 8, and 11 (education-num, relationship, and capital-gain respectively) were extracted into census_col.

5.4 Male Employees in State Government

Filtering for male employees whose workclass is 'State-gov' yielded 795 matching records, stored in male_gov.

5.5 39-Year-Olds with a Bachelor's Degree or US Nativity

Filtering for individuals aged exactly 39 who either hold a Bachelor's degree or are natives of the United States (an inclusive OR condition) yielded 742 matching records, stored in census_us.

5.6 Random Sample of 200 Rows

A random sample of 200 rows was drawn from the cleaned dataset (with a fixed random seed for reproducibility) and stored in census_200, useful for quick manual inspection or lightweight testing.

5.7 Count of Workclass Categories

The distribution of individuals across the different workclass categories is as follows:

Workclass	Count
Private	22,286
Self-emp-not-inc	2,499
Local-gov	2,067
State-gov	1,279
Self-emp-inc	1,074
Federal-gov	943
Without-pay	14

The overwhelming majority of the workforce (about 74%) is employed in the private sector.

5.8 Average Capital Gain by Workclass

Grouping by workclass and computing the mean of capital-gain reveals which employment categories are associated with the highest investment income:

Workclass	Mean Capital Gain (\$)
Self-emp-inc	4,810.75
Self-emp-not-inc	1,913.13
Private	879.86
Federal-gov	832.32
Local-gov	829.23
State-gov	684.31
Without-pay	487.86

Self-employed individuals (particularly those with incorporated businesses) show substantially higher average capital gains than salaried employees, which is consistent with business ownership generating investment-style income.

5.9 High Earners by Gender

Separate subsets were created for males and females earning more than \$50,000 annually. The male subset contains 6,396 records, while the female subset contains 1,112 records. This large disparity (roughly 85% male vs. 15% female among high earners) reflects gender-based workforce and income patterns present in the 1994 census data.

5.10 Percentage of US Private-Sector Employees Earning $\leq$ \$50,000

Among individuals who are natives of the United States, work in the private sector, and earn \$50,000 or less annually, the percentage relative to the entire cleaned dataset is 51.70%. This single group therefore makes up more than half of the entire dataset.

5.11 Percentage of Married Individuals

Treating any marital-status value containing the word 'Married' (covering civilian spouse, spouse-absent, and armed-forces spouse categories) as married, 80.17% of the dataset is composed of married individuals.

5.12 Percentage of High School Graduates Earning More Than \$50,000

Among individuals whose highest education level is 'HS-grad', 16.43% earn more than \$50,000 annually. Expressed as a percentage of the entire dataset, high school graduates earning above \$50,000 make up 5.36% of all records. This is noticeably lower than the overall high-earner rate, suggesting that a high school education alone is associated with a lower likelihood of high income compared to higher education levels (explored further in Section 6).

6. Exploratory Data Analysis (EDA)

Visual exploration was carried out to better understand feature distributions and their relationship with the income target variable.

6.1 Age Distribution

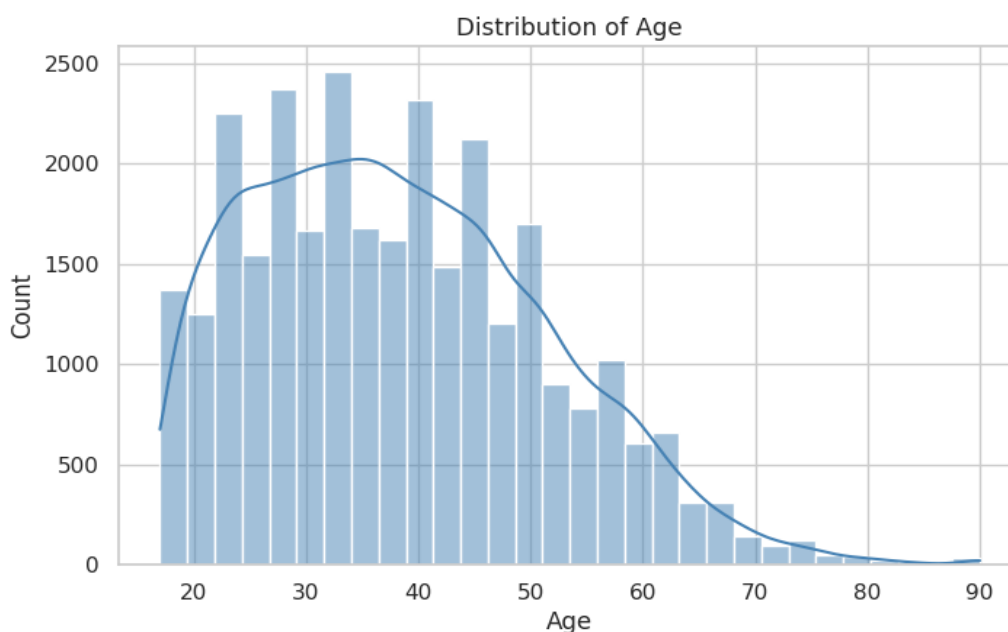


Figure 1: Distribution of age across the dataset.

The age distribution is right-skewed, with most individuals concentrated between 25 and 50 years old, and a long tail of older workers extending toward 90 years.

6.2 Income Class Distribution



Figure 2: Count of individuals earning $\leq \$50K$ versus $> \$50K$.

The dataset is imbalanced: a considerably larger proportion of individuals earn $\leq \$50K$ than $> \$50K$. This imbalance is important to keep in mind when interpreting classification accuracy, since a naive model that always predicts ' $\leq \$50K$ ' would already achieve a moderately high accuracy.

6.3 Workclass Distribution

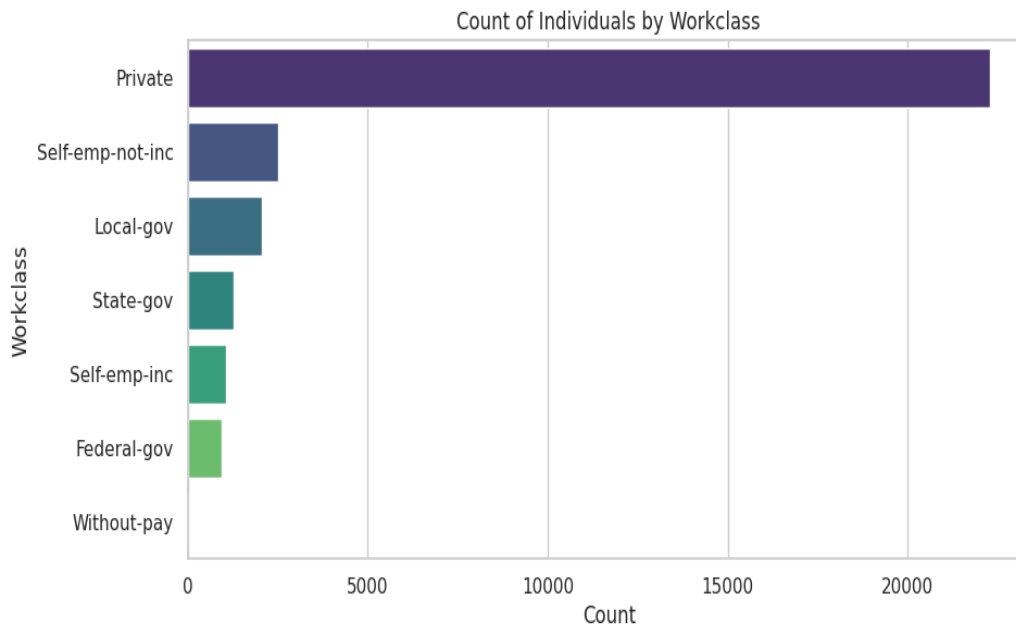


Figure 3: Number of individuals in each workclass category.

As noted in Section 5.7, the Private sector dominates the workforce, followed by self-employment and government roles.

6.4 Hours per Week vs. Income

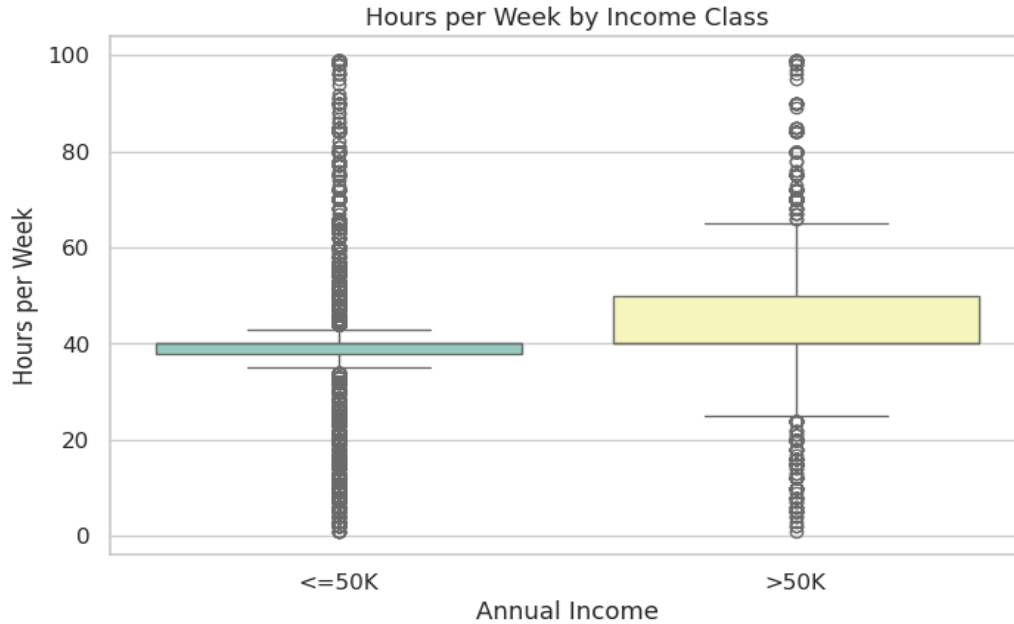


Figure 4: Distribution of weekly working hours split by income class.

Individuals earning $> \$50K$ tend to work more hours per week on average, with a higher median and a narrower interquartile range compared to individuals earning $\leq \$50K$.

6.5 Education vs. Income

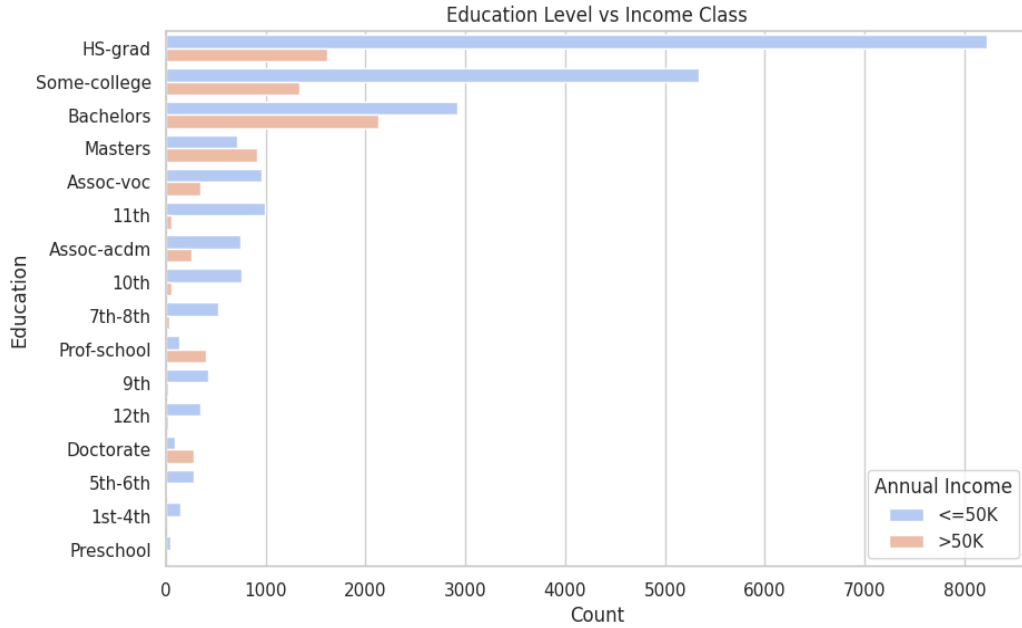


Figure 5: Education level versus income class.

Higher education levels (Bachelors, Masters, Doctorate, Prof-school) show a markedly higher proportion of individuals earning >\$50K, confirming education as one of the strongest predictors of income in this dataset.

6.6 Correlation Heatmap

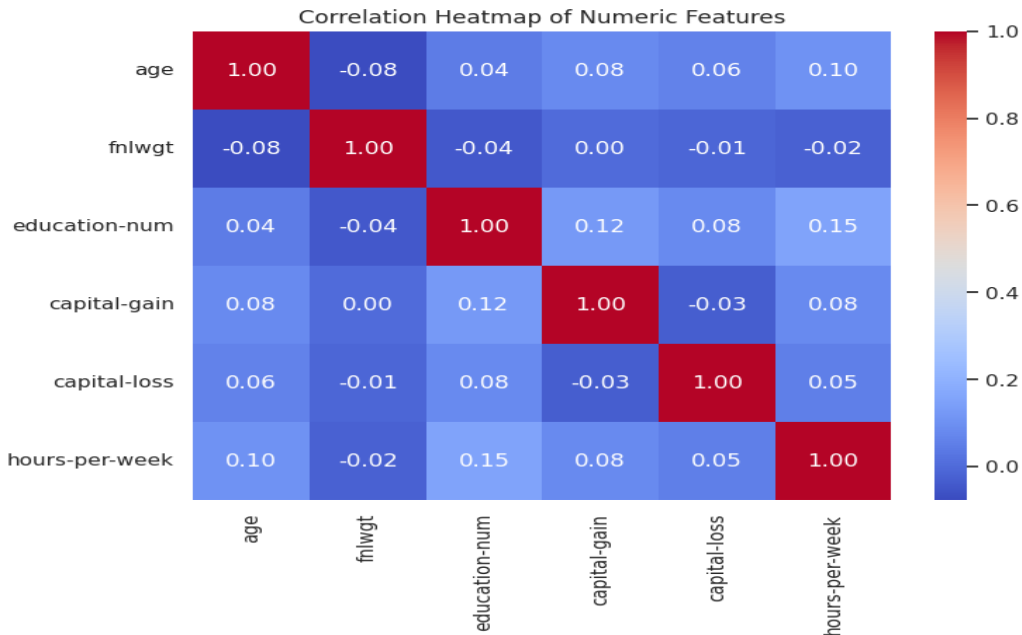


Figure 6: Correlation matrix of numeric features.

Correlations among the numeric features are generally weak, indicating limited multicollinearity. Education-num shows the strongest (though still modest) positive correlation with hours-per-week and capital-gain, both of which align with intuitive expectations about income-related behavior.

7. Machine Learning Methodology

Five models were built in total, covering both regression and classification tasks. For classification, the target variable annual_income was encoded numerically as a new column X, where 0 represents '≤\$50K' and 1 represents '>\$50K'. Categorical predictors were encoded using either one-hot (dummy) encoding for logistic regression models, or label encoding for the tree-based models, which naturally handle encoded categorical splits.

Model	Target	Predictors	Train:Test Split
Linear Regression	hours-per-week	education-num	70 : 30
Logistic Regression (Simple)	X (income class)	occupation	65 : 35
Logistic Regression (Multiple)	X (income class)	age, workclass, education	80 : 20
Decision Tree	X (income class)	All remaining features	70 : 30
Random Forest (300 trees)	X (income class)	All remaining features	80 : 20

8. Linear Regression: Education vs. Hours Worked

A simple linear regression model was built with education-num as the independent variable and hours-per-week as the dependent variable, using a 70:30 train-test split.

The model produced the following equation:

$$\text{hours-per-week} = 34.06 + 0.68 \times \text{education-num}$$

This suggests that, on average, each additional unit of education level is associated with approximately 0.68 additional hours worked per week.

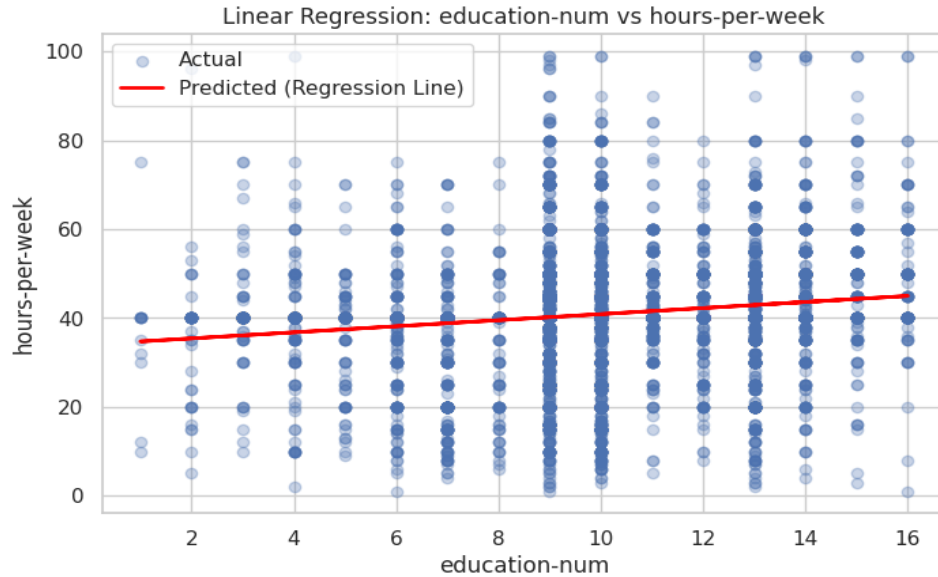


Figure 7: Linear regression fit of education-num against hours-per-week on the test set.

Metric	Test Set	Training Set
Mean Absolute Error (MAE)	7.52	7.51
Mean Squared Error (MSE)	137.81	141.20
Root Mean Squared Error (RMSE)	11.74	11.88
R ² Score	0.028	—

The very low R² value (0.028) indicates that education level alone explains only about 2.8% of the variability in weekly working hours. This is expected, since working hours are influenced by many factors beyond education — such as occupation type, employment status, and personal circumstances — that are not included in this single-variable model. The similar error metrics between the training and test sets indicate the model is not overfitting; it is simply too simple to capture the underlying relationship well.

9. Logistic Regression: Simple Model (Occupation Only)

This model predicts income class using only the occupation attribute (one-hot encoded), split 65:35 into training and test sets.

	Predicted ≤\$50K	Predicted >\$50K
Actual ≤\$50K	7,929	0
Actual >\$50K	2,628	0

Accuracy: 75.11%

The confusion matrix reveals an important limitation of this simple model: it never predicts the '>\$50K' class at all. Because occupation alone does not provide enough separating power, and because the dataset is imbalanced toward the '≤\$50K' class, the model defaults to always predicting the majority class. This produces a respectable-looking accuracy of 75.11% purely because roughly three-quarters of the dataset falls into that majority class — not because the model has learned a meaningful decision boundary. This is a classic illustration of why accuracy alone can be a misleading metric on imbalanced data, and why metrics such as precision, recall, and F1-score (or techniques like class weighting) are valuable complements in real-world use.

10. Logistic Regression: Multiple Model (Age, Workclass, Education)

This model predicts income class using age, workclass, and education as predictors, split 80:20 into training and test sets.

	Predicted ≤\$50K	Predicted >\$50K
Actual ≤\$50K	4,280	251
Actual >\$50K	1,045	457

Accuracy: 78.52%

Adding age, workclass, and education as predictors improves accuracy over the occupation-only model, from 75.11% to 78.52%. Unlike the simple model, this model does successfully identify a meaningful number of true positives (457 correctly identified high earners), indicating that education level and age together carry real predictive signal for income class.

11. Decision Tree Model

A Decision Tree classifier was trained using all available features (label-encoded), split 70:30 into training and test sets.

	Predicted ≤\$50K	Predicted >\$50K
Actual ≤\$50K	5,920	877
Actual >\$50K	834	1,418

Accuracy: 81.09%

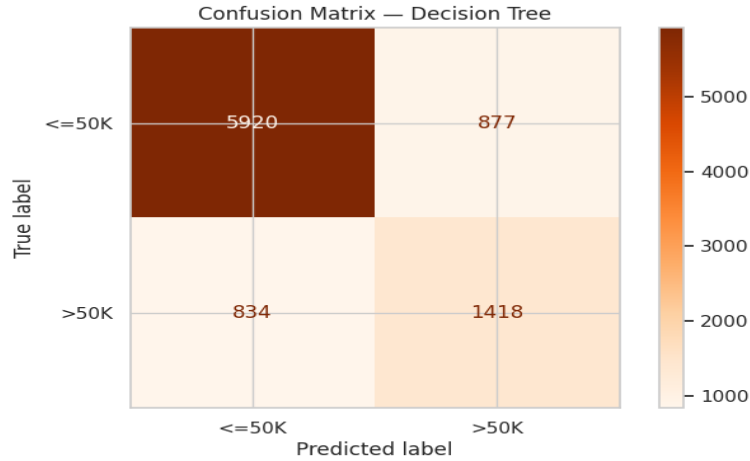


Figure 8: Confusion matrix — Decision Tree.

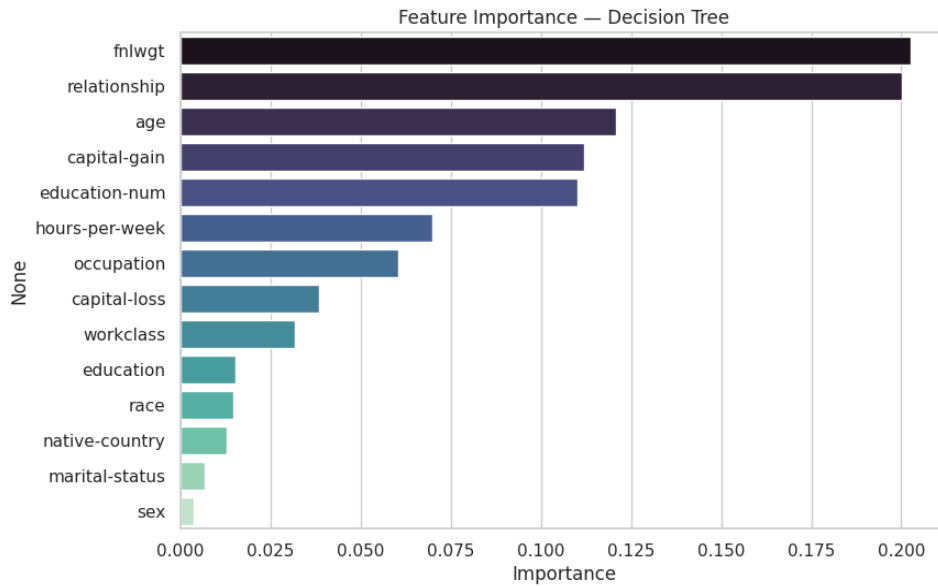


Figure 9: Feature importance scores from the Decision Tree.

Using the full feature set substantially improves performance over the logistic regression models, reaching 81.09% accuracy. The feature importance chart shows which attributes the tree relied on most heavily when making its splits — typically features such as relationship status, marital status, education, capital-gain, and age tend to dominate in this dataset, reflecting their strong association with income level.

12. Random Forest Model (300 Trees)

A Random Forest classifier with 300 trees was trained using the same full feature set as the Decision Tree, split 80:20 into training and test sets.

	Predicted $\leq \$50K$	Predicted $> \$50K$
Actual $\leq \$50K$	4,202	329
Actual $> \$50K$	551	951

Accuracy: 85.41%

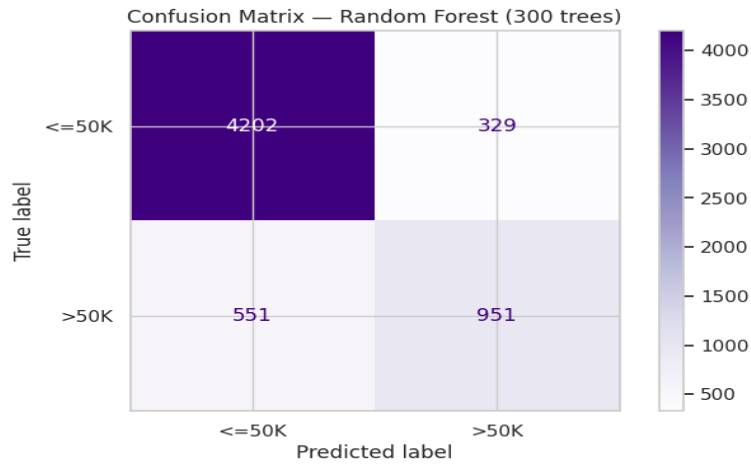


Figure 10: Confusion matrix — Random Forest (300 trees).

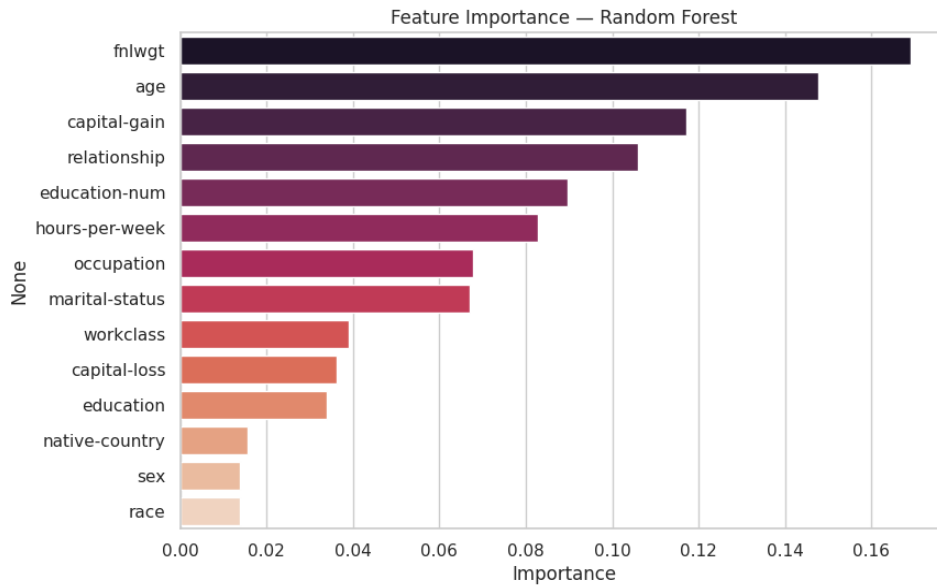


Figure 11: Feature importance scores from the Random Forest.

The Random Forest achieves the best performance of all models tested, at 85.41% accuracy. By averaging predictions across 300 individual decision trees, each trained on a different bootstrap sample and random subset of features, the model reduces the overfitting and variance that a single Decision Tree is prone to, resulting in more stable and generalizable predictions.

13. Model Comparison and Discussion

Rank	Model	Accuracy
1	Random Forest (all features, 300 trees)	85.41%
2	Decision Tree (all features)	81.09%
3	Logistic Regression (age, workclass, education)	78.52%
4	Logistic Regression (occupation only)	75.11%

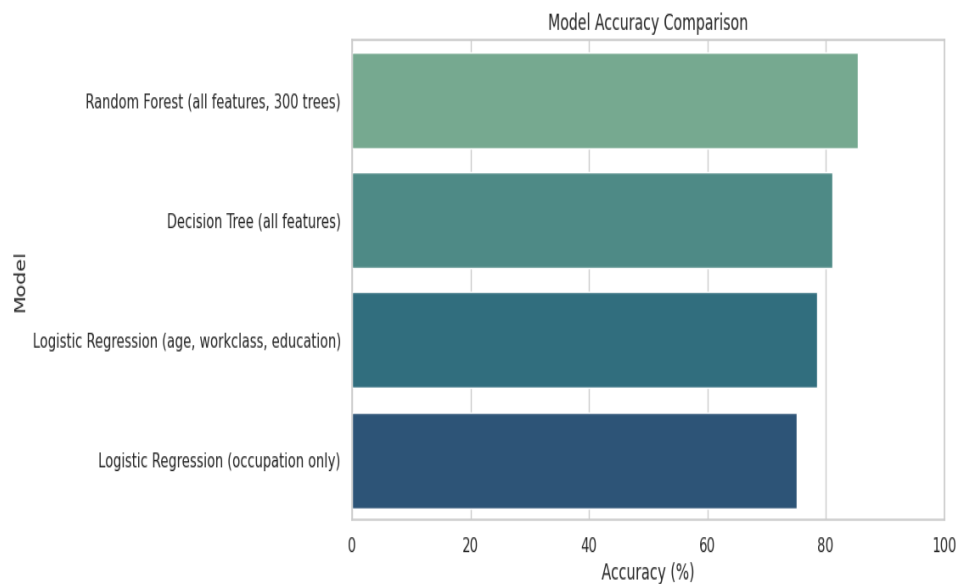


Figure 12: Accuracy comparison across all classification models.

The results follow a clear and expected pattern. Model accuracy improves as (a) more relevant features are included and (b) the model's capacity to capture non-linear relationships increases. The simple logistic regression, limited to a single categorical feature, performs the weakest and fails to predict the minority class at all. Adding more features to the logistic regression measurably improves results. Moving to tree-based models that can use every available feature and capture non-linear interactions produces a further significant jump in accuracy, with the Random Forest — an ensemble of many trees — outperforming a single Decision Tree by reducing variance and overfitting.

It is worth noting that all models still show room for improvement in correctly identifying the minority '>\$50K' class, which is common in imbalanced classification problems. This is discussed further in the recommendations below.

14. Conclusion

This project successfully demonstrates a complete data science pipeline on the Census Income dataset: data cleaning, structured data manipulation, exploratory visualization, and predictive modeling. Key findings include:

- Roughly 7.4% of records contained missing data (marked as '?'), concentrated in the workclass, occupation, and native-country columns.
- Education level, hours worked per week, and marital/relationship status show strong visual associations with income class.
- The private sector employs about 74% of the workforce, and self-employed individuals show notably higher average capital gains.
- 80.17% of individuals in the dataset are married, and married status is strongly associated with the '>\$50K' income group (largely reflecting dual-income households and the demographic profile of the 'Husband' relationship category).
- Among classification models, Random Forest (300 trees) achieved the best performance at 85.41% accuracy, followed by Decision Tree (81.09%), multiple Logistic Regression (78.52%), and simple Logistic Regression (75.11%).
- The simple, single-feature logistic regression model illustrates the risk of relying on accuracy alone when working with imbalanced datasets, since it achieved a seemingly reasonable accuracy while never predicting the minority class.

15. Limitations and Future Scope

15.1 Limitations

- Class imbalance (roughly 3:1 in favor of '≤\$50K') affects model evaluation; accuracy alone can overstate real-world usefulness.
- Listwise deletion of missing values, while simple, discards information rather than imputing it, and may introduce mild bias if missingness is not completely random.
- Label encoding of categorical variables for tree-based models imposes an artificial ordinal relationship on unordered categories, which trees can partially work around through repeated splits but is not ideal in principle.
- The dataset reflects 1994 US labor market conditions and demographic patterns, which may not generalize to the present day.

15.2 Recommendations for Future Work

- Apply class-imbalance handling techniques such as class weighting, oversampling (e.g., SMOTE), or threshold tuning to improve recall on the minority '>\$50K' class.
- Evaluate models using precision, recall, F1-score, and ROC-AUC in addition to accuracy for a fuller picture of performance.
- Tune hyperparameters (e.g., tree depth, minimum samples per leaf, number of estimators, regularization strength) using cross-validation and grid or random search.

- Experiment with gradient boosting methods (e.g., XGBoost, LightGBM, Gradient Boosting) which often outperform Random Forest on tabular data.
- Consider feature engineering, such as combining capital-gain and capital-loss into net capital change, or bucketing age and hours-per-week into meaningful ranges.

16. References and Tools

- Dataset: Census Income (“Adult”) Dataset — originally sourced from the UCI Machine Learning Repository, based on 1994 US Census Bureau data.
- Python Software Foundation — Python Language Reference.
- Pandas Development Team — pandas documentation.
- Pedregosa et al. — Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research.
- Hunter, J.D. — Matplotlib: A 2D Graphics Environment.
- Waskom, M. — Seaborn: statistical data visualization.